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In-vitro antibacterial effect of medicinal plant (*Gymnema sylvestre*) extract.

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Abstract : The antibacterial activities of Gymnema sylvestre[1] were selected . The main aim of the research work is to confirm the ethno medicinal claim that Gymnema Sylvestre has got antibacterial activities which could be a better alternative for synthetic antimicrobials, if proved to be effective enough[2]. For this we have extracted Gymnema Sylvestre with Petroleum Ether then with Chloroform and lastlty with metanol, using Soxhlet Apparatus. Antibacterial activity of different extracts were evaluated by "Paper Disc Diffusion Method"[3] against Bacillus subtilis, Staphylococcus aureus and Escherichia Coli using Gentamicin as a standard and DMSO as Control Sample. Zone of inhibition of Bacillus Subtilis and Staphylococcus Aureus were found respectively when compared with the result of the Standard, while Extracts had no activity against E.Coli. The described method is a rapid, homogenous , cost effective and easy to perform system for screening of new potential antimicrobial agents in drug discovery[3]. So, in future the leaves of this plant can be used for selective infections and can be further explored for isolation of the active principles responsible for antimicrobial property.[6]

Key Words: *Gymnema Sylvestre*, Paper disc Diffusion Method, Gentamicin, Zone of Inhibition

INTRODUCTION

Gymnema sylvestre is an important medicinal plant native to Deccan Peninsula is also found in different parts of Chota Nagpur pleateau region, commonly known as Gurmar. *Gymnema sylvestre* R.Br. (Asclepiadaceae) commonly known as "Gurmar" (Destroyer of sugar) leaves have been used for centuries in the traditional Indian system of Ayurvedic medicine[9]. The term "destroyer of sugar"[1] is traditionally used for Gymnema because chewing the leaves will abolish the taste of sweetness[11]. That is, sweet foods no longer tasted sweet, but rather became almost completely tasteless. The medicinally active parts of the plant are the leaves and the roots. Gymnema sylvestre(also called gurmar),a woody plant that grows wild in India,has a long history of use in Ayurvedic medicine. The leaves of G. sylvestre contain glycosides, the peptide gurmarin, gymnemic acids , saponins, stigmasterol, quercitol, and several amino acid derivatives. It is used in the treatment of Diabetes Mellitus and in Food Additives against Obesity and carries. The Powder of Gymnema Root has also been used in Snake Bites, Constipation, Stomach Complaints, Water Retension and Liver diseases[20]. It is also found to be used against Infections in Traditional Systems of Medicine. Some works have already been done for its antimicrobial property, but we are trying the same using Leaf extracts using different Solvents for extraction and to find out which fraction is more effective.

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MATERIALS AND METHODS

Materials required :

1. The leaves of Gymnema Sylvestre were collected from Mother plant ,Jamshedpur.
2. Soxhlet Apparatus.
3. Pathological Strains i.e. , Staphylococcus Aureus (Gram Positive) and Escherichia Coli (Gram Negative).

4. Nutrient Agar Media .
5. Standard Drug: Gentamicin discs
6. DMSO
7. Paper discs were prepared from Whattmann Filter Paper.

METHEDOLOGY:

1. The collected Plant Material (Leaf) was shade dried at room temp. then they were pulverized in Mixer Grinder and passed through sieve of 60 mesh size.
2. Accurately weighed 500gm of Powdered Leaves and Extracted with different Solvents Choosing from Non Polar to Polar(Petroleum Ether >Chloform>metanol) using Soxhlet Apparatus.
3. The different extracts were used in different concentrations to evaluate the antimicrobial properties using Paper Disc Diffusion Method.
4. Various concentrations of the extracts were prepared (i.e.10ìg/ml, 50ìg/ml, 100ìg/ml & 200ìg/ml) with DMSO.
5. Dip the paper discs in these concentrations of extract for some time.
6. Sterilized medium was inoculated with suspension of microorganisms and poured into Petri dishes to give a depth of 3 to 4 mm.
7. The paper discs impregnated with tested extracts were placed on solidified mediums.
8. Then the standard disc of Gentamicin was placed on the same medium at different locations. Similarly placed the disc of control Sample (DMSO) on the same medium at different locations.
9. The plates were pre incubated for 1 hr. at room temperature and then incubated ay 37°c for 24 hrs. for determining antibacterial activity.
10. The observed zone of inhibition is presented in the table below.

Table: Zone of Inhibition in different concentration (ìg/ml)

Extracts	Concentrations In µg/ml	Area of zone of inhibition			
		Bacillus subtillis	Staphylococcus aureus	Escherichi a coli	Standard (Gentamicin)
Petroleum Ether	10	-	-	-	4
	50	7	8	-	10
	100	10	9	-	12
	200	20	16	-	22
Chloroform	10	-	-	-	6
	50	8	7	-	11
	100	10	10	-	16
	200	20	19	-	25
Metanol	10	-	-	-	6
	50	10	9	-	14
	100	12	12	-	18
	200	24	20	-	25

Note: Control Sample(DMSO) has not shown any Zone of Inhibition.

RESULTS AND DISCUSSION :

Different Extracts isolated from Gymnema Sylvestre were found to have Significant antibacterial activity against Gram Positive bacteria and no activity was seen against Gram Negative bacteria as compared to standard drug Gentamicin. Antibacterial activity was evaluated by Paper Disc Diffusion method against Bacillus Subtilis, Staphylococcous aureus, E.Coli using Gentamicine as a Standard Drug and DMSO as a control sample. Larger and remarkable antimicrobial activities were recorded with Bacillus Subtilis, Staphylococcous aureus.

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